RESTRICTING A GROUPOID SCHEME OVER A FIELD TO A FIELD EXTENSION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A morphism with target the spectrum of a field is universally open (Morphisms chapter, Lemma morphisms-lemma-scheme-over-field-universally-open, canonical tag 0383). cited at proof lines 1244-1245. Gives that s, t : R to U and the extension morphism U' to U are universally open, the open half of the assertion.
- Surjective morphisms are stable under base change (Morphisms chapter, Lemma morphisms-lemma-base-change-surjective, canonical tag 01S1). cited at proof lines 1248-1249. Transfers the surjectivity of s, t and of U' to U to the base changed arrows of the diagram.
- Compositions of surjective morphisms are surjective (Morphisms chapter, Lemma morphisms-lemma-composition-surjective, canonical tag 01S0). cited at proof lines 1248-1250. Handles the composite arrows of the diagram, such as the long arrows R' to U and R' to U' through R.
- Compositions of universally open morphisms are universally open (Morphisms chapter, Lemma morphisms-lemma-composition-open, canonical tag 02V2). cited at proof lines 1249-1251. Combines the universal openness of U' to U and of s, t on the composite arrows of the diagram.
- Affine morphisms are stable under base change (Morphisms chapter, Lemma morphisms-lemma-base-change-affine, canonical tag 01SD). cited at proof lines 1251-1252. Makes the dotted arrow R' to R, a base change of the affine U' to U, affine; likewise the other base changed arrows.
- Compositions of affine morphisms are affine (Morphisms chapter, Lemma morphisms-lemma-composition-affine, canonical tag 01SC). cited at proof lines 1251-1253. Covers the composite affine arrows of the diagram.
- Flat morphisms are stable under base change (Morphisms chapter, Lemma morphisms-lemma-base-change-flat, canonical tag 01U9). cited at proof lines 1253-1254. Transfers flatness of s, t and of U' to U to the base changed arrows of the diagram.

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- Compositions of flat morphisms are flat (Morphisms chapter, Lemma morphisms-lemma-composition-flat, canonical tag 01U7). cited at proof lines 1253-1255. Covers the composite flat arrows of the diagram, completing the verification for all arrows.

Source reference. Original lemma 0383: sources/stacks-chow/tags.txt, source lines 4222–4222. Cited at proof lines 1244-1245: s, t : R to U and U' to U are universally open.

Source reference. Original lemma 01S1: sources/stacks-chow/tags.txt, source lines 2322–2322. Cited at proof lines 1248-1249: surjectivity is stable under the base changes occurring in the defining diagram.

Source reference. Original lemma 01S0: sources/stacks-chow/tags.txt, source lines 2321–2321. Cited at proof lines 1248-1250: surjectivity passes to the composite arrows of the diagram.

Source reference. Original lemma 02V2: sources/stacks-chow/tags.txt, source lines 3753–3753. Cited at proof lines 1249-1251: universal openness passes to the composite arrows of the diagram.

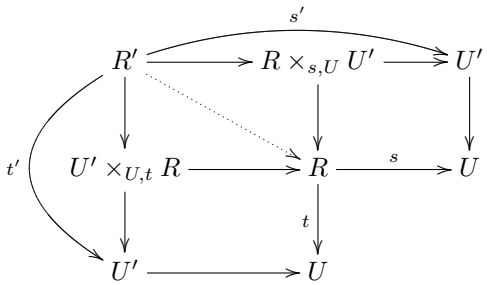
Source reference. Original lemma 01SD: sources/stacks-chow/tags.txt, source lines 2334–2334. Cited at proof lines 1251-1252: the dotted arrow R' to R, a base change of the affine U' to U, is affine.

Source reference. Original lemma 01SC: sources/stacks-chow/tags.txt, source lines 2333–2333. Cited at proof lines 1251-1253: affineness passes to the composite arrows of the diagram.

Source reference. Original lemma 01U9: sources/stacks-chow/tags.txt, source lines 2402–2402. Cited at proof lines 1253-1254: flatness is stable under the base changes occurring in the defining diagram.

Source reference. Original lemma 01U7: sources/stacks-chow/tags.txt, source lines 2400–2400. Cited at proof lines 1253-1255: flatness passes to the composite arrows of the diagram.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume $U = \mathrm{Spec}(k)$ with k a field. Let k'/k be a field extension, $U' = \mathrm{Spec}(k')$ and let (U', R', s', t', c') be the restriction of (U, R, s, t, c) via $U' \to U$. In the defining diagram*



all the morphisms are surjective, flat, and universally open. The dotted arrow $R' \to R$ is in addition affine.

Proof. The morphism $U' \to U$ equals $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$, hence is affine, surjective and flat. The morphisms $s, t : R \to U$ and the morphism $U' \to U$ are universally open by Morphisms, Lemma 0383. Since R is not empty and U is the spectrum of a field the morphisms $s, t : R \to U$ are surjective and flat. Then you conclude by using Morphisms, Lemmas 01S1, 01S0, 02V2, 01SD, 01SC, 01U9, and 01U7. □